

Selenium Java AWS Interview Prep

so i have interview with client for selenium java testig, they have asked me to preparpe aws as well.. what are the things i should b e knowing.. its selenium java profile....

For a **Selenium + Java Automation** interview, when they say "**prepare AWS as well**", they usually do **not** expect a DevOps engineer level of AWS knowledge. They want to know how automation testing fits into cloud environments.

Focus on these topics:

Topic	What You Should Know
AWS Basics	What is AWS, Region, Availability Zone, VPC
EC2	Launching Windows/Linux VM, why used for Selenium Grid
S3	Object storage, storing reports/screenshots/logs
IAM	Users, Roles, Policies, least privilege concept
CloudWatch	Monitoring logs and metrics
Lambda	Serverless functions, triggering automation tasks
RDS	Managed databases (MySQL/Postgres)
ECS/EKS	Running Docker containers in cloud
CodePipeline / CodeBuild	CI/CD concepts
Auto Scaling	Adding/removing servers automatically

Topic	What You Should Know
Security Groups	AWS firewall concept

Most Important Selenium-related AWS Questions

1. Why use AWS in Test Automation?

Example answer:

AWS provides scalable infrastructure where Selenium tests can run on EC2 instances or Docker containers. Test reports can be stored in S3 and execution can be integrated with CI/CD pipelines.

2. What is EC2?

EC2 (Elastic Compute Cloud) is a virtual machine service in AWS.

Interview follow-up:

How did you use EC2?

Answer:

We hosted Selenium Grid on EC2 instances and executed automation suites remotely.

3. What is S3?

S3 is object storage.

Common use cases:

- Store Extent Reports
 - Store screenshots
 - Store logs
 - Backup test data
-

4. What is IAM?

Very common question.

IAM manages authentication and authorization in AWS.

Know:

- User
- Group
- Role
- Policy

Example:

Jenkins running on EC2 can assume an IAM role to upload reports to S3.

5. Difference between EC2 and Lambda?

EC2	Lambda
Server required	Serverless
Long-running	Short execution
Manage OS	AWS manages infrastructure

6. How would you run Selenium tests in AWS?

Typical architecture:

```
Jenkins
|
v
EC2 Selenium Grid
|
v
Execute Tests
|
v
Store Reports in S3
```

7. Docker + AWS

Very popular nowadays.

Know:

Selenium Grid

- > Docker Containers
- > AWS ECS/EKS

If they ask:

Why Docker?

Answer:

Consistent environment, easier browser version management, faster parallel execution.

8. What is CloudWatch?

Monitoring service for logs and metrics.

Examples:

- Application logs
 - CPU utilization
 - Memory alerts
 - Failure notifications
-

9. What is VPC?

Virtual Private Cloud is an isolated network inside AWS.

Know these words:

- Public Subnet
- Private Subnet
- Internet Gateway

No need for deep networking unless the role is SDET/DevOps-heavy.

Questions a Selenium Interviewer Might Ask

How can Selenium tests be executed in AWS?

- EC2
- Docker on ECS
- Kubernetes on EKS

- Lambda (not common for UI tests)
-

How would you store screenshots generated by Selenium?

Upload screenshots to S3 using AWS SDK.

How would you trigger test execution automatically?

Jenkins → CodePipeline → EC2 Selenium Grid.

What AWS service would you use to store Extent Reports?

S3.

What AWS service would you use for monitoring failed executions?

CloudWatch.

For a 12-year Selenium Automation profile

I would prioritize:

1. EC2
2. S3
3. IAM
4. CloudWatch
5. Docker + ECS
6. Jenkins integration with AWS
7. Selenium Grid on AWS
8. Basic VPC concepts

These 8 topics cover about **80–90% of AWS questions** that appear in Selenium Automation interviews. If you want, I can also give you ↪ [20 AWS interview questions with model answers specifically targeted at Selenium/SDet interviews.](#)

sure ..give it to me

Here are 20 AWS questions with Selenium/SDET-focused answers that are realistic for a 10–12+ years Selenium Java Automation interview.

1. What is AWS?

Answer:

AWS (Amazon Web Services) is a cloud platform that provides services such as virtual servers, storage, databases, networking, monitoring, and CI/CD tools.

In automation projects, AWS is commonly used to host Selenium Grid, Jenkins, applications under test, and test reports.

2. What is EC2?

Answer:

EC2 (Elastic Compute Cloud) provides virtual machines in the cloud.

Common uses:

- Host Selenium Grid
 - Host Jenkins
 - Run automation suites
 - Execute parallel browser tests
-

3. Why use EC2 instead of local machines?

Answer:

- Scalable
- Available 24/7
- Easier team access
- Can create multiple test environments
- Supports parallel execution

Example:

Instead of running tests on my laptop, I can run them on multiple EC2 instances simultaneously.

4. What is S3?

Answer:

S3 is AWS object storage.

In testing:

- Store screenshots
 - Store Extent Reports
 - Store execution logs
 - Store test data files
-

5. Difference between EBS and S3?

EBS	S3
Attached to EC2	Independent storage
Block storage	Object storage
Acts like hard disk	Acts like file storage

Example:

Selenium Grid VM uses EBS. Reports are stored in S3.

6. What is IAM?

Answer:

IAM controls who can access AWS resources.

Components:

- Users
 - Groups
 - Roles
 - Policies
-

7. What is an IAM Role?

Answer:

A role provides temporary permissions.

Example:

Jenkins running on EC2 assumes an IAM role to upload reports to S3 without storing access keys.

8. What is a Security Group?

Answer:

Security Group acts as a firewall for EC2.

Example:

```
Allow:  
22  -> SSH  
80  -> HTTP  
443 -> HTTPS  
4444 -> Selenium Grid
```

9. What is a VPC?

Answer:

VPC (Virtual Private Cloud) is a private network in AWS.

It provides:

- Isolation
- Security
- Controlled access

10. What is the difference between Public and Private Subnet?

Public	Private
Internet access	No direct internet
Web servers	Databases
Selenium Grid UI	Backend services

11. What is CloudWatch?

Answer:

CloudWatch is AWS monitoring service.

Used for:

- CPU monitoring
 - Memory monitoring
 - Log monitoring
 - Alerting
-

12. How can CloudWatch help testing?

Answer:

CloudWatch can alert when:

- Selenium Grid crashes
 - EC2 CPU becomes high
 - Disk space becomes low
 - Application logs show errors
-

13. What is AWS Lambda?

Answer:

Lambda is a serverless compute service.

You upload code and AWS executes it on demand.

No server management required.

14. Can Selenium tests run in Lambda?

Answer:

Possible but uncommon.

Reasons:

- Execution time limits
- Browser dependency issues
- Memory limitations

Most UI tests run on EC2, ECS, or EKS.

15. What is Docker?

Answer:

Docker packages application and dependencies into containers.

Benefits:

- Consistent environment
 - Easier browser version management
 - Faster deployment
-

16. Why use Docker with Selenium?

Answer:

Instead of installing browsers manually:

```
Chrome Container  
Firefox Container  
Edge Container
```

Each container has its own browser.

This makes parallel execution easier.

17. What is ECS?

Answer:

ECS (Elastic Container Service) is AWS's container orchestration service.

Used to run Docker containers without managing many servers.

Example:

Selenium Docker containers can run on ECS.

18. What is EKS?

Answer:

EKS is managed Kubernetes in AWS.

Used for:

- Large-scale container deployments
- Auto-scaling test infrastructure

If your company uses Kubernetes, know this term.

19. How would you execute Selenium tests in AWS?

A typical answer:

```
Developer Commit
  |
  v
Jenkins
  |
  v
EC2/ECS Selenium Grid
  |
  v
Execute Tests
  |
  v
Reports in S3
  |
  v
Alerts through CloudWatch
```

This answer impresses interviewers because it shows end-to-end understanding.

20. Describe a Selenium Automation Framework on AWS

A strong senior-level answer:

Our Selenium Java TestNG framework was triggered through Jenkins. Tests executed on Selenium Grid running on EC2/Docker containers. Test reports and screenshots were uploaded to S3. CloudWatch monitored infrastructure health. IAM roles were used for secure access to AWS resources.

Bonus: One-Liners Interviewers Love

What is Auto Scaling?

Automatically increases or decreases EC2 instances based on load.

What is Load Balancer?

Distributes traffic across multiple servers.

What is Route 53?

AWS DNS service.

What is RDS?

Managed relational database service.

What is CodePipeline?

AWS CI/CD orchestration service.

What is CodeBuild?

AWS build and test execution service.

For your Selenium Java profile, if you can confidently explain:

- Selenium Grid
- Jenkins
- Docker
- EC2
- S3
- IAM

- CloudWatch
- ECS/EKS (high level)

you'll sound like someone who has worked with cloud-based test automation, even if your day-to-day work has mainly been Selenium Java + TestNG.